

Preface

Numerical Analysis

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Chapter 1

Preliminaries

§ 1.1. Big "Oh" and Little "Oh" Notation

§ 1.2. Norms

Chapter 2

Finite Difference Approximations

§ 2.1. Method of Undetermined Coefficients

Given a smooth function u and a point $\bar{x}$, we wish to approximate $u^{(n)}(\bar{x})$ using only the neighboring points of $\bar{x}$ which differ by multiples of some (small) positive h . The method of undetermined coefficients is an algorithm which does this.

Example 2.1.1. Let:

$$u''(\bar{x}) = A \cdot u(\bar{x} - h) + B \cdot u(\bar{x}) + C \cdot u(\bar{x} + h) + D \cdot u(\bar{x} + 2h). \quad (2.1)$$

Note that the Taylor series of $u(x)$ about the point $\bar{x}$ is:

$$u(x) = \sum_{n=0}^{\infty} \frac{u^{(n)}(\bar{x})}{n!} (x - \bar{x}).$$

By evaluating this Taylor expansion at $\bar{x} - h$, $\bar{x} + h$, and $\bar{x} + 2h$, we can rewrite Equation (1) as:

$$u''(\bar{x}) = A \sum_{n=0}^{\infty} \frac{u^{(n)}(\bar{x})}{n!} (x - \bar{x})^n + B \sum_{n=0}^{\infty} \frac{u^{(n)}(\bar{x})}{n!} (x - \bar{x})^n + C \sum_{n=0}^{\infty} \frac{u^{(n)}(\bar{x})}{n!} (x - \bar{x})^n + D \sum_{n=0}^{\infty} \frac{u^{(n)}(\bar{x})}{n!} (x - \bar{x})^n$$